costantino.pala.geo@proton.me @
Italy

Costantino Pala

Geospatial Software Developer & Geohazard Analyst

R&D Geospatial Developer and Licensed Geologist with a PhD in Earth and Environmental Sciences. I combine field expertise in post-wildfire geohazard assessment with end-to-end software development — designing EO pipelines, GIS frameworks, and spatial analysis tools that translate complex satellite and terrain datasets into operational outputs. My work spans from sediment connectivity modelling and burn severity mapping to building SUBSTR8 and INUE, a Sentinel-2-powered emergency response tool mapping 700 km² at 10 m resolution in under 15 minutes.

GEOSPATIAL SKILLS

- QGIS
- QField
- ArcGIS Pro
- Geoprocessing
- Spatial Analysis
- Data Management
- ArcPy
- Python
- SUBSTR8 (developer)

PROFESSIONAL EXPERIENCE

Doctoral Researcher - GIS Developer — **November 2022 – February 2026**

University of Cagliari

- Designed and developed INUE (AGPL-3.0), an automated post-wildfire EO pipeline built on SUBSTR8, combining Sentinel-2 imagery (dNBR, NDVI) with DEM-derived sediment connectivity indices (TauDEM/MPI) to generate erosion susceptibility maps across 700 km² at 10 m resolution in under 15 minutes for emergency response operations.
- Optimized infrastructure costs by creating SUBSTR8, a high-performance GIS platform that executes complex spatial workflows on legacy hardware.
- Translated complex datasets into strategic insights for executive-level audiences while managing end-to-end project lifecycles and technical documentation.

GIS Engineering Intern — **October 2023 – January 2024**

CSIC-IGME - Madrid, Spain

- Modelled post-wildfire rockfall hazard to characterize debris source areas for cascading debris flow risk assessment
- Integrated thermal spalling analysis into a multi-hazard (rockfall → debris flow) framework for burned catchments

Geospatial Research Associate — **April 2022 - October 2022**

University of Cagliari, Italy

- Mapped post-wildfire erosional and depositional landforms through systematic field survey
- Produced soil and colluvium thickness distribution maps across the burned area
- Built a geospatial database of landslides and mass movements within the fire-affected catchment

LANGUAGES

English	Proficient, C1
French	Upper-Intermediate, B2
Spanish	Upper-Intermediate, B2
Italian	Native
Sardinian	Native

SOFTWARE DEVELOPED

- SUBSTR8: Modular Python framework for building standardized geospatial analysis software. Engineered for hardware-agnostic performance via dynamic chunking and multithreading (Dask/NumPy/Rasterio), achieving 0.0085 ms/pixel on low-end hardware. Provides six core modules — raster I/O, GUI, file management, mapping, and parameter handling — as a reusable foundation for custom spatial pipelines and standalone executables.

- INUE: SUBSTR8-powered automated EO pipeline for post-wildfire erosion susceptibility mapping. Processes Sentinel-2 multispectral imagery (NIR, SWIR, Red) and DEM data to compute the FIREX index — integrating burn severity (dNBR), sediment connectivity (IC), and geomorphic disconnection factors — mapping 700 km^2 at 10 m resolution in under 15 minutes to support emergency response prioritization.

EDUCATION

PhD in Earth and Environmental Sciences and Technologies — **October 2022 - February 2026**
University of Cagliari

Licensed Geologist — **June 2021**
University of Cagliari

Degree in Geological Sciences and Technologies — **September 2018 - July 2020**
University of Cagliari

OTHER TECHNICAL SKILLS

OS: Windows, Linux and Android. **Programming:** Python and software development
Productivity & Design: Microsoft Office Suite, Open Source Office Suite, Inkscape, MS Teams
Command Line: shell and bash

ADDITIONAL SKILLS

- Driving License (Category B) and own vehicle
- Assistance in drafting proposals for European Union project calls

16 April 2026